



School of Information
144 West 14th Street, 6th floor
New York, NY 10011-7301

INFO 656: Machine Learning

Professor Filipa Calado

Fall 2026

Course Information

Section: 01

Semester: Fall 2026

Meeting Dates & Times: Wednesdays, 3:00 PM - 5:50 PM

Credits: 3

Prerequisites or other restrictions: 664

Instructor Contact Information

Filipa Calado, PhD (she)

Office Location: Pratt Manhattan, room 602

Office Hours: Mondays 5:00pm - 6:00pm, Wednesdays 1:30pm-2:30pm, and by appointment at: <https://calendly.com/fcalado-pratt/new-meeting>

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Course Description:

Machine learning is a rapidly growing field that develops algorithms for tasks such as data classification and prediction. These algorithms are programmed to operate and adjust themselves independently of human intervention (i.e., to learn), allowing data work to occur quickly and at scale. Machine learning is a key technology behind the automation across many social areas today, often branded AI.

This course offers an introduction to machine learning as a practical tool that we can use, and as a technological field with social implications. We will learn about key concepts in machine learning; survey a few key machine learning techniques, such as supervised methods for machine learning (regression and classification), which attempt to map data onto desired outputs, and unsupervised methods (clustering and association), which attempt to find structure within data itself; use openly available tools to implement these techniques on text and image data; learn how to assess the effectiveness of different techniques on particular datasets; and discuss basic issues that confront all machine learning methods.

Detailed Description:

This course prioritizes a critical approach to machine learning and recent advances in ML research. Therefore, in addition to learning the basics of ML principles and models, this course centers issues from ML research, including dataset curation, sustainability, algorithmic bias, and auditing & evaluation methods, among others.

At a time when automated coding assistants provide much technical knowledge (i.e., applying algorithms and writing program code) this critical approach will endow students with crucial human-centered skills for developing and critiquing ML tools.

Course Goals:

The goals of this course are to:

- introduce students to key concepts and some common techniques in machine learning, as well as openly available tools
- help students to develop technical and critical thinking skills regarding machine learning
- enable students to conduct a machine learning experiment and communicate the result of their project

Student Learning Outcomes:

By the end of the course, students will be able to:

- describe different machine learning methods, including their limitations
- select an appropriate machine learning method for a given use case
- implement machine learning algorithms and assess their performance
- execute a machine learning experiment using openly available tools
- support the design of their experiment by discussing both the technical aspect and the topic's significance

Course Requirements

Sources and Materials

Class notebooks are available at: <https://github.com/gofilipa/656/tree/main/class-notebooks>

Assignments

Participation in class discussion. 20% of final grade. Because we spend much of the course learning about concepts from ML, including recent research, you must do the readings and be prepared to discuss them in class, as well as to offer your opinion. Some

of the weekly homeworks are designed to get you thinking about the readings (see below about the "provocations" homework). You are encouraged to share your provocations during class discussion.

Weekly homeworks. 25% of final grade. These assignments may be coding assignments or they may be short written responses (i.e. "provocations"), approximately 1 page or 300 words in length. Coding assignments will be based on skills that we learned in class the previous day, and the instructions will be described in detail in the course schedule. Written responses, also called "provocations," consist of a focused reflection on the reading, and must include a question meant to provoke discussion among peers (a provocative, thoughtful, and open-ended question). You may use the provocation as an opportunity to express, work through, and develop your thoughts regarding an idea or quote from the reading, with the goal of eventually arriving at a question for further discussion.

Final project. 25% of final grade. A machine learning project of your choosing, such as a Small Language Model fine-tuned on custom data, or a Sentiment Analysis experiment. You might use ML techniques that we learned in class, but can also go beyond these techniques. All data, methodology, and code should be well annotated. Finally, project must include a README.md explaining the main components (specific guidelines will be distributed in the middle of the semester). Can be done in partners but if that's the case the final project should scale in complexity. Includes a final presentation between 5 - 10 minutes long (the exact length for the presentation is TBD – it will depend on how many individuals vs groups end up presenting).

Conference paper. 30% of final grade. 8 - 12 pages in length. This is a group project. The paper should explore a current issue in ML research and propose an intervention in the form of an experiment, audit, or review. It should be written in accordance with the Call For Papers for the FAccT conference: <https://facctconference.org/2027/cfp.html>

Assessment & Grading

Grade Breakdown

Participation in class discussion: 20%

Weekly homeworks. 25%

Final project: 25%

Conference paper: 30%

Attendance alone is not considered a percentage of the grade, but the overall grade may be lowered based on absences (see Attendance Policy below).

Pratt Institute Grade System

Grade Evaluation Quality Points per Credit

A Excellent 4.0

A- Excellent 3.7

B+ Very Good 3.3

B Good 3.0

B- Good 2.7

C+ Above Average 2.3

C Average 2.0

C- Below Average (UG only) 1.7

D+ Less than Acceptable (UG only) 1.3

D Less than Acceptable (UG only) 1.0

F Failure 0.0

WF Failure Due to Lack of Attendance 0.0

Course Schedule

Unit 1: Foundational Algorithms in ML

Week 1 (Aug 26): Introduction

Readings (to be done in class):

- HAI article, "AI Hiring Tools Can Yield Racial Bias and Systemic Rejection." <https://hai.stanford.edu/news/ai-hiring-tools-can-yield-racial-bias-and-systemic-rejection>.

Activities:

- Part 1: introductions; syllabus review
- Part 2: read & discuss HAI article; python refresher

Week 2 (Sept 2): Data & Bias

Readings:

- Abeba Birhane, Sepehr Dehdashtian, Vinay Prabhu, and Vishnu Boddeti. 2024. "The Dark Side of Dataset Scaling: Evaluating Racial Classification in Multimodal Models." In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, June 03, 2024. ACM, Rio de Janeiro Brazil, 1229–1244. <https://doi.org/10.1145/3630106.3658968>
- Dodge, Jesse, et al. "Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus." *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, edited by Marie-Francine Moens et al., Association for Computational Linguistics, 2021, pp. 1286–305. ACLWeb, <https://doi.org/10.18653/v1/2021.emnlp-main.98>.

Homework #2 due:

- write a provocation on the "Dataset Scaling" essay 1 - 2 pages double spaced. Submit the provocation on Canvas, and be prepared to share your ideas verbally with the class.

Activities:

- Part 1: share provocations and discuss readings; begin python lab on data collections and cleaning with pandas using data from the MET API.
- Part 2: continue python lab; data exploration and cleaning in groups.

Week 3 (Sept 9): Supervised Learning: Linear & Logistic Regression

Readings:

- Andrew Ng's video explaining Linear Regression, watch until minute 42: https://www.youtube.com/watch?v=4b4MUYve_U8&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU&index=2 (or, link to the non-scary version: https://www.youtube.com/watch?v=dLc-lfEEYss&list=PLkDaE6sCZn6FNC6YRfRQc_FbeQrF8BwGI&index=9)
- explore Chris Deotte, "Classifier Playground": <http://www.ccom.ucsd.edu/~cdeotte/programs/classify.html>

Homework #3 due:

- find a dataset on NYC Open Data, Kaggle, or elsewhere. Open it in a python notebook and practice cleaning and/or structuring the dataset. Write up a summary of how you cleaned the data, as a comment in the last cell. Submit your notebook to canvas.

Activities:

- Part 1: discussion of linear regression; in class python lab on linear regression with scikit-learn
- Part 2: discussion of logistic regression; in class python lab on logistic regression with scikit-learn

Week 4 (Sept 16): Unsupervised Learning: K-means Clustering & Decision Trees

Readings:

- Lavigne, Sam, et al. "Predicting Financial Crime: Augmenting the Predictive Policing Arsenal." arXiv:1704.07826, arXiv, 25 Apr. 2017. arXiv.org, <https://doi.org/10.48550/arXiv.1704.07826>.

Homework #4 due:

- find a dataset to practice either linear or logistic regression using sci-kit learn. In the last cell of the notebook, write up a summary of what you learned about your data and/or topic, and what potential blind spots, oversights, or risks might be associated with your analysis. Submit your notebook on canvas.

Activities:

- Part 1: python lab on k-means clustering
- Part 2: discuss White Collar Crime project

Week 5 (Sept 23): Model Evaluation

Readings:

- Devinney, Hannah, et al. “We Don’t Talk About That: Case Studies on Intersectional Analysis of Social Bias in Large Language Models.” Proceedings of the 5th Workshop on Gender Bias in Natural Language Processing (GeBNLP), edited by Agnieszka Faleńska et al., Association for Computational Linguistics, 2024, pp. 33–44. ACLWeb, <https://doi.org/10.18653/v1/2024.gebnlp-1.3>.
- Sui, Peiqi, et al. “Critical Confabulation: Can LLMs Hallucinate for Social Good?” arXiv:2511.07722, arXiv, 6 Mar. 2026. arXiv.org, <https://doi.org/10.48550/arXiv.2511.07722>.

Homework #5 due:

- Do some research on current issues in ML. You might check out some of the papers from this summer’s FAccT conference: <https://facctconference.org/2026/acceptedpapers.html>. Pick **one** recent paper that seem interesting to you. Write up a short summary, and explain why you find it interesting. Submit the summaries on Canvas, and be ready to briefly share and discuss them with the class.

Activities:

- Part 1: discuss evaluations, practice hands-on with evaluation software
- Part 2: share out papers that we are interested in so far; class discussion on topics; organize into provisional teams for conference paper.

Unit 2: ML Research: Conference Papers

Week 6 (Sept 30): Conference paper workshop day I

Readings:

- None (except for homework)

Homework #6:

- With your team, submit a research question (there can be more than one), a list of potential sources, and a detailed work plan for the next month, with distribution of tasks and expected time allocated for each team member.

Activities:

- Part 1: discuss annotated bibliographies
- Part 2: share research questions & come up with check-in schedule for the next month

Week 7 (Oct 7): Conference paper workshop day II

Readings:

- None (except for homework)

Homework #7 due:

- Submit an annotated bibliography of at least 12-20 papers (4-5 papers per group member) that are relevant to your team's topic(s) of interest. 4-5 sentences of annotation per paper, including at least one sentence explaining how this paper may be useful for further ML research on the topic.

Activities:

- Part 1: discuss outlining & the parts of an academic paper
- Part 2: group work on papers

Week 8 (Oct 14): Conference paper workshop day III

Readings:

- *How to Write a Computer Science Research Paper: Complete Step-by-Step Guide*. <https://smartacademicwriting.com/how-write-computer-science-research-paper/>

Homework #8 due:

- Submit a detailed outline for the first draft, which includes all of the major sections of a typical CS paper. In your outline, include topics and/or claims, and key evidence/quotations.

Activities:

- Part 1: discuss "Abstract," "Results" & "Discussion" sections.
- part 1: group work on paper

Week 9 (Oct 21): Conference paper workshop day IV (abstracts due)

Note: Abstracts are due on FAccT website on Monday, Oct 27, if submitting.

Readings:

- Filipa Calado. 2025. Some Myths About Bias: A Queer Studies Reading Of Gender Bias In NLP. In Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP), pages 338–346, Vienna, Austria. Association for Computational Linguistics. <https://aclanthology.org/2025.gebnlp-1.29/>
- Some Myths About Bias, peer review. <https://github.com/gofilipa/656/blob/main/peer-review-gebnlp.pdf>

Homework #9 due:

- Submit a draft of the Abstract, Results and/or Discussion sections of the paper.

Activities:

- Part 1: peer review of abstracts, Results, and Discussion sections
- Part 2: discuss & plan revisions

Week 10 (Oct 28): Conference paper workshop day V (drafts due)

Note: final papers due on FAccT on Tuesday, November 3, if submitting

Readings:

- TBD

Homework #10:

- Submit full draft of paper.

Activities:

- Part 1: peer review of full paper
- Part 2: discuss & plan revisions

Unit 3: Language Modeling & Final Projects

Week 11 (Nov 4): Word Vectors

Readings:

- Jay Alammam, "The Illustrated Word2Vec" 2018. <https://jalammar.github.io/illustrated-word2vec/>
- Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient Estimation of Word Representations in Vector Space. <https://doi.org/10.48550/arXiv.1301.3781>

Homework:

- None

Activities:

- Part 1: discuss readings on word embeddings; python lab on Word2Vec
- Part 1: continue Word2Vec; individual practice

Week 12 (Nov 11): Transformer models & Backprop

Readings:

- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. "Attention Is All You Need." <https://doi.org/10.48550/arXiv.1706.03762>
- Jay Alammam. The Illustrated BERT, ELMo, and co. (How NLP Cracked Transfer Learning). Retrieved April 14, 2025 from <https://jalammar.github.io/illustrated-bert/>

- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. "Language Models are Unsupervised Multitask Learners." https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
- Optional (maybe helpful!):
 - 3Blue1Brown, "Neural Networks" YouTube playlist: https://www.youtube.com/playlist?list=PLZHQ0b0WTQDNU6R1_67000Dx_ZCJB-3pi
 - Chris Deotte, "Neural Network Playground": <http://www.ccom.ucsd.edu/~cdeotte/programs/neuralnetwork.htm>

Activities:

- Part 1: discuss Transformer architecture; python lab on backprop (Karpathy's version)
- Part 2: continue python lab

Week 13 (Nov 18): Fine-Tuning

Readings:

- Sarah Ciston. A CRITICAL FIELD GUIDE FOR WORKING WITH MACHINE LEARNING DATASETS. <https://knowingmachines.org/critical-field-guide>. Accessed 22 July 2026.
- Bots, LLMs, and Automated Access — Internet Archive Developer Portal. <https://archive.org/developers/bots.html>. Accessed 8 July 2026.
- FineWeb: Decanting the Web for the Finest Text Data at Scale - a Hugging Face Space by HuggingFaceFW. <https://huggingface.co/spaces/HuggingFaceFW/blogpost-fineweb-v1>. Accessed 7 July 2026.

Activities:

- Part 1: python lab on fine-tuning
- Part 2: python lab on web scraping

week 14 (Nov 25): NO CLASS - thanksgiving break

week 15 (Dec 2): Project workshops

Readings:

- None

Homework #15:

- Submit a README.md file for your final project

Activities:

- Part 1: work on projects
- Part 2: work on projects

week 16 (Dec 9): Presentations

Course Resources

Cool ML projects

- Thenewinquiry/Whitecollar. 26 Feb. 2017, Python. The New Inquiry, 19 Oct. 2021. GitHub, <https://github.com/thenewinquiry/whitecollar>.
- “Mr. Chatterbox, or, The Modern Prometheus.” Estragon, 30 Mar. 2026, <https://www.estrakon.news/mr-chatterbox-or-the-modern-prometheus/>.
- “Developing Emotion-Based Speech Technology for the Māori Language.” Data & Society, <https://datasociety.net/points/developing-emotion-based-speech-technology-for-the-maori-language/>. Accessed 7 May 2026.

Sample Datasets

- “Immigration and Customs Enforcement” data. Collected by the Deportation Data Project. <https://deportationdata.org/data/processed/ice.html>
- “LGBTQ+ Legislative Tracking 2025.” Google Docs, https://docs.google.com/spreadsheets/u/1/d/1NfY8VvrUNQKX77CXilx1E1pepRSGTMZ70LoyFwx6fbk/edit?gid=719771929&usp=embed_facebook. Accessed 8 July 2026.

Learning resources and tutorials (my personal recommendations)

- Andrej Karpathy’s *Microgpt* project. <https://karpathy.github.io/2026/02/12/microgpt/>.
- Andrew Ng’s playlist of the CS229 course at Stanford: <https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShTMGgzqpfVfbU>
- Andrew Ng’s playlist of Machine Learning course on Coursera: https://www.youtube.com/playlist?list=PLkDaE6sCZn6FNC6YRfRQc_FbeQrF8BwGI
- Language Modelling — The Complete Guide — Language Modelling — The Complete Guide. <https://karam-nus.github.io/language-modelling/>.
- py4u blog. “Python Web Scraping: Why Requests Triggers Cloudflare Security (But Urllib Doesn’t) – Solutions Explained.” Py4u, December 1, 2025. <https://www.py4u.org/blog/python-s-requests-triggers-cloudflare-s-security-while-urllib-does-not/>.

Policies

The following abbreviated set of policies is especially relevant to this class. Full details on policies and procedures can be found on the Pratt website or in the Office of the Vice President for Student Affairs, Main Building, Lower Level.

Community Standards

All Pratt students, faculty, and staff members are expected to value and uphold the community standards essential to the pursuit of academic excellence and social responsibility. These include expectations for social conduct, academic integrity, non-discrimination, and other policies described in the link above, and apply to all Pratt-sponsored activities, on or off campus.

Academic Integrity

Academic integrity at Pratt means using your own and original ideas in creating academic work. It also means that if you use the ideas or influence of others in your work, you must acknowledge them.

At Pratt,

- We do our own work,
- We are creative, and
- We give credit where it is due.

When students submit any work for academic credit, they make an implicit claim that the work is wholly their own, completed without the assistance of any unauthorized person. These works include, but are not limited to exams, quizzes, presentations, papers, projects, studio work, and other assignments and assessments. In addition, no student shall prevent another student from making their work. Students may study, collaborate, and work together on assignments at the discretion of the instructor.

Examples of infractions include but are not limited to:

The following examples are drawn from the Academic Integrity Code and should be curated or supplemented based on assignments for your course.

1. Plagiarism, defined as using the exact language or a close paraphrase of someone else's ideas without citation.
2. Violations of fair use, including the unauthorized and uncited use of another's artworks, images, designs, etc.
3. The supplying or receiving of completed work including papers, projects, outlines, artworks, designs, prototypes, models, or research for submission by any person other than the author.
4. The unauthorized submission of the same or essentially the same piece of work for credit in two different classes.
5. The unauthorized supplying or receiving of information about the form or content of an examination.
6. The supplying or receiving of partial or complete answers, or suggestions for answers; or the supplying or receiving of assistance in interpretation of questions on any examination from any source not explicitly authorized. (This includes copying or reading of another student's work or consultation of notes or other sources during an examination.)

7. The use of generative artificial intelligence (AI) to produce or to improve work, whether visual or textual, except when called for by an assignment or instructor and acknowledged transparently as one tool among others in the creative process.

The Academic Integrity Standing Committee (AISC) is charged with educating faculty, staff, and students about academic integrity practices. Whenever possible, we strive to resolve alleged infractions at the most local level possible, such as between student and professor, or within a department or school. When necessary, members of this committee will form an Academic Integrity Hearing Board to hear cases regarding cheating, plagiarism, and other infractions described below; these infractions can be grounds for citation, sanction, or dismissal. Detailed procedures are explained in the full version of the Academic Integrity Policy.

Attendance Policy

Consistent attendance is essential for the completion of any course or program. Attending class does not earn students any specific portion of their grade, but is the pre-condition for passing the course, while missing class may seriously harm a student's grade. Pratt Institute respects students' requirements to observe days of cultural significance, including religious holy days, and recognizes that some students might need to miss class to do so. In this, or other similar, circumstance, students are responsible for consulting with faculty ahead of time about how and when they can make up work they will miss. Reasonable accommodations for students with documented disabilities will continue to be provided, as appropriate. For full details, see Pratt's Attendance Policy.

The expectations for this class are as follows:

- Come to every class
- Arrive promptly
- If for some reason you cannot make it to class, or will be late to class, please email me and let me know ahead of time.

Those who email me ahead of time with a good reason for missing class will incur an excused absence. Those who do not will incur an unexcused absence. For each unexcused absence, grades may be lowered a letter grade. Three or more unexcused absences in some courses (especially those that meet only once per week) may result in an automatic "F" for the course.

Academic Support

Multiple academic support resources are available to students across campus:

- For assistance with time management and/or studio, subject, and software tutoring, contact the Student Success Center at success@pratt.edu.
- For assistance with writing assignments, contact the Writing and Tutorial Center at wtc@pratt.edu. The Pratt Libraries can also help with research and citations.
- Academic advisors are also a great resource; students can find their advisor's contact information or schedule an appointment through Starfish.

Accessibility

The Learning/Access Center (L/AC) coordinates access for students with disabilities. Students who identify as having any type of disability are entitled and encouraged to enroll with the L/AC in order to determine and implement reasonable accommodations. Contact the Learning/Access Center at lac@pratt.edu or 718.802.3123 for information or to schedule an appointment.

Title IX and Sexual Misconduct Policy

Pratt Institute is committed to fostering an environment that is safe, secure and free from sex discrimination and sexual harassment, sexual violence, dating and domestic violence, and stalking among all forms of sexual misconduct. The Institute takes prompt and appropriate action to address misconduct, end a hostile environment if one has been created, and prevent the recurrence of a hostile environment. To submit a concern, please use the Title IX and Sexual Misconduct Disclosure Form. For full details, see the Title IX and Sexual Misconduct Policy.

If you inform me of an issue of sex discrimination or sexual misconduct, I will keep the information as private as I can. However, as a faculty member, I am a mandatory reporter of sexual misconduct and required to bring it to the attention of the Institute's Title IX Coordinator, who can also be contacted at titleix@pratt.edu. You can also speak to someone confidentially by contacting our confidential resources in Health Services or the Counseling Center.

Nondiscrimination and Anti-Harassment Policy

Pratt is committed to fostering a welcoming, safe, non-discriminatory, and harassment-free educational, living, and working environment for its community. To submit a concern, please use the Discrimination and Harassment Disclosure Form, which also allows anonymous reports. For full details, see the Nondiscrimination and Anti-Harassment Policy.

Wellbeing

Pratt is dedicated to creating a culture where the entire community can flourish and thrive. Taking time to care for yourself and seeking appropriate support is important to achieving your academic and professional goals. Several resources are available through Starfish, including our Student Advocate and Care Coordinator, who can also be reached at 718.399.4546 or studentadvocate@pratt.edu. If you or anyone you know experiences overwhelming academic stress, persistent difficult feelings, or challenging life events, the Counseling Center can be reached 24/7 by calling 718.687.5356. To schedule a consultation, please call or email therapy@pratt.edu

Starfish

Starfish provides students with contact information for advisors, instructors, department chairs, and support services across the campus. Faculty can also use Starfish to inform

students of their progress in class and connect them with resources. Students are contacted by support offices whenever a flag is raised.

Public Safety & Emergency Contacts

The Department of Public Safety provides 24-hour-a-day protection to the campus, including an emergency response guide. Contact Public Safety at security@pratt.edu or 718.636.3540. The Pratt Emergency Alert System is used to send urgent messages to registered mobile devices and emails (faculty, staff, and students have the option to opt-out, which must be renewed each year). You can update your emergency contact information in the Safety section of OnePratt.